Year 10 Higher

# REVISION PACK

## Number 1

**Name:** _____

**Class:** _____

## Number 1 · Working with fractions (Lesson 1.1)

**Worked example: adding fractions**

Work out $\frac{3}{4} + \frac{5}{6}$

Give your answer as a mixed number.

| | | |
|---|---|---|
| **Step 1** | Same bottom | $\frac{9}{12} + \frac{10}{12}$ |
| **Step 2** | Add the tops | $\frac{19}{12}$ |
| **Step 3** | Answer | $1\frac{7}{12}$ |

### Practice: adding fractions

**1.1** Work out $\frac{4}{5} + \frac{2}{3}$. Give your answer as a mixed number.

**1.2** Work out $\frac{5}{6} + \frac{3}{8}$. Give your answer as a mixed number.

**1.3** Work out $\frac{3}{4} + \frac{2}{3}$. Give your answer as a mixed number.

**1.4** Work out $\frac{5}{7} + \frac{1}{2}$. Give your answer as a mixed number.

**1.5** Work out $\frac{7}{8} + \frac{1}{3}$. Give your answer as a mixed number.

**1.6** Work out $\frac{5}{9} + \frac{5}{6}$. Give your answer as a mixed number.

## Number 1 · Working with fractions (Lesson 1.1)

**Worked example: multiplying mixed numbers**

Work out $1\frac{2}{3} \times 2\frac{1}{4}$

Give your answer as a mixed number in its simplest form.

| | |
|---|---|
| **Step 1** Improper fractions | $\frac{5}{3} \times \frac{9}{4}$ |
| **Step 2** Multiply | $\frac{45}{12}$ |
| **Step 3** Simplify | $\frac{15}{4}$ |
| **Step 4** Answer | $3\frac{3}{4}$ |

### Practice: multiplying mixed numbers

**1.7** Work out $1\frac{1}{2} \times 2\frac{2}{3}$

**1.8** Work out $3\frac{1}{3} \times 1\frac{3}{5}$. Give your answer as a mixed number in its simplest form.

**1.9** Work out $2\frac{1}{4} \times 1\frac{5}{6}$. Give your answer as a mixed number in its simplest form.

**1.10** Work out $1\frac{3}{4} \times 2\frac{2}{5}$. Give your answer as a mixed number in its simplest form.

**1.11** Work out $2\frac{2}{3} \times 1\frac{1}{8}$

**1.12** Work out $3\frac{3}{4} \times 1\frac{2}{3}$. Give your answer as a mixed number in its simplest form.

## Number 1 · Working with fractions (Lesson 1.1)

**Worked example: dividing mixed numbers**

Work out $2\frac{1}{4} \div 1\frac{1}{2}$

| | |
|---|---|
| **Step 1** Improper fractions | $\frac{9}{4} \div \frac{3}{2}$ |
| **Step 2** Flip and multiply | $\frac{9}{4} \times \frac{2}{3} = \frac{18}{12}$ |
| **Step 3** Simplify | $\frac{3}{2}$ |
| **Step 4** Answer | $1\frac{1}{2}$ |

### Practice: dividing mixed numbers

**1.13** Work out $4\frac{1}{2} \div 1\frac{1}{5}$. Give your answer as a mixed number in its simplest form.

**1.14** Work out $2\frac{2}{3} \div 1\frac{1}{3}$

**1.15** Work out $3\frac{3}{4} \div 2\frac{1}{2}$. Give your answer as a mixed number in its simplest form.

**1.16** Work out $5\frac{1}{4} \div 1\frac{3}{4}$

**1.17** Work out $2\frac{2}{5} \div 1\frac{1}{5}$

**1.18** Work out $4\frac{1}{6} \div 1\frac{2}{3}$. Give your answer as a mixed number in its simplest form.

## Number 1 · Working with fractions (Lesson 1.1)

**Worked example: subtracting with a mixed number**

Work out $2\frac{1}{4} - \frac{2}{3}$

| | |
|---|---|
| **Step 1** Improper fraction | $\frac{9}{4} - \frac{2}{3}$ |
| **Step 2** Same bottom | $\frac{27}{12} - \frac{8}{12}$ |
| **Step 3** Subtract the tops | $\frac{19}{12}$ |
| **Step 4** Answer | $1\frac{7}{12}$ |

### Practice: subtracting with a mixed number

**1.19** Work out $2\frac{1}{3} - \frac{5}{6}$

**1.20** Work out $1\frac{3}{5} - \frac{3}{4}$

**1.21** Work out $3\frac{1}{4} - \frac{2}{3}$

**1.22** Work out $2\frac{2}{5} - \frac{5}{6}$

**1.23** Work out $1\frac{5}{6} - \frac{3}{4}$

**1.24** Work out $2\frac{1}{2} - \frac{4}{5}$

## Number 1 · Working with fractions (Lesson 1.1)

**Worked example: fraction problems**

Sara spends $\frac{1}{3}$ of the money on rent. Sara spends $\frac{1}{4}$ of the rest on food. What fraction of the money does Sara have left?

| | |
|---|---|
| **Step 1** The rest | $1 - \frac{1}{3} = \frac{2}{3}$ |
| **Step 2** Food | $\frac{1}{4} \times \frac{2}{3} = \frac{1}{6}$ |
| **Step 3** Left | $\frac{2}{3} - \frac{1}{6} = \frac{1}{2}$ |
| **Step 4** Answer | $\frac{1}{2}$ |

### Practice: fraction problems

**1.25** Omar spends $\frac{2}{5}$ of the money on rent. Omar spends $\frac{1}{3}$ of the rest on food. What fraction of the money does Omar have left?

**1.26** Huda spends $\frac{1}{4}$ of the money on clothes. Huda spends $\frac{2}{3}$ of the rest on books. What fraction of the money does Huda have left?

**1.27** Yousef spends $\frac{3}{8}$ of the money on a phone. Yousef spends $\frac{1}{5}$ of the rest on games. What fraction of the money does Yousef have left?

**1.28** Mona spends $\frac{1}{6}$ of the money on food. Mona spends $\frac{3}{5}$ of the rest on travel. What fraction of the money does Mona have left?

**1.29** Fahad spends $\frac{3}{7}$ of the money on rent. Fahad spends $\frac{1}{2}$ of the rest on food. What fraction of the money does Fahad have left?

**1.30** Laila spends $\frac{2}{9}$ of the money on books. Laila spends $\frac{3}{7}$ of the rest on clothes. What fraction of the money does Laila have left?

## Number 1 · Order of operations and rounding (Lessons 1.2, 1.3)

**Worked example: order of operations**

This question is about the order of operations.

(a) Work out $2 + 4 \times 3^2$

(b) Insert one pair of brackets to make this calculation correct.
$8 + 2 \times 5 - 1 = 49$

| | | | |
|---|---|---|---|
| (a) | Step 1 | Indices first | $3^2 = 9$ |
| | Step 2 | Then multiply | $4 \times 9 = 36$ |
| | Step 3 | Then add | $2 + 36 = 38$ |
| (b) | Step 1 | Brackets round 8 + 2 | $(8 + 2) \times 5 - 1$ |
| | Step 2 | Check | $10 \times 5 - 1 = 49$ |

### Practice: order of operations

**2.1** This question is about the order of operations.

(a) Work out $7 + 2 \times 5^2$

(b) Insert one pair of brackets to make this calculation correct. $3 + 5 \times 4 - 2 = 30$

**2.2** This question is about the order of operations.

(a) Work out $20 - 3 \times 2^3$

(b) Insert one pair of brackets to make this calculation correct. $2 \times 6 + 3 - 1 = 17$

**2.3** This question is about the order of operations.

(a) Work out $6 + 18 \div 3^2$

(b) Insert one pair of brackets to make this calculation correct. $4 + 8 \times 2 + 1 = 28$

**2.4** This question is about the order of operations.

(a) Work out $9 + 4 \times 2^3$

(b) Insert one pair of brackets to make this calculation correct. $5 \times 4 - 1 + 2 = 17$

**2.5** This question is about the order of operations.

(a) Work out $3 + 6 \times 2^2$

(b) Insert one pair of brackets to make this calculation correct. $10 - 2 \times 3 + 1 = 25$

**2.6** This question is about the order of operations.

(a) Work out $40 - 5 \times 3^2$

(b) Insert one pair of brackets to make this calculation correct. $6 + 2 \times 7 - 3 = 53$

## Number 1 · Order of operations and rounding (Lessons 1.2, 1.3)

**Worked example: rounding**

Round 562.738

(a) correct to 1 decimal place. (b) correct to 2 significant figures.

| | | | |
|---|---|---|---|
| **(a)** | **Step 1** | Next digit is 3: round down | $562.7$ |
| **(b)** | **Step 1** | Next digit is 2: round down | $560$ |

### Practice: rounding

**2.7** Round 275.846

(a) correct to 1 decimal place. (b) correct to 2 significant figures.

**2.8** Round 1093.472

(a) correct to 1 decimal place. (b) correct to 2 significant figures.

**2.9** Round 48.0961

(a) correct to 1 decimal place. (b) correct to 2 significant figures.

**2.10** Round 736.258

(a) correct to 1 decimal place. (b) correct to 2 significant figures.

**2.11** Round 5281.63

(a) correct to 1 decimal place. (b) correct to 2 significant figures.

**2.12** Round 19.954

(a) correct to 1 decimal place. (b) correct to 2 significant figures.